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RESEARCH ARTICLE

A Dynamic Phishing URL Detection System Integrating Advanced Machine Learning and Crowdsourced Threat Intelligence

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ABSTRACT Phishing remains a critical and evolving threat in the cybersecurity landscape, consistently exploiting human factors to bypass technical defenses. This paper proposes a dynamic phishing detection framework that integrates Artificial Intelligence (AI) and crowdsourcing to improve detection performance and adaptability. The proposed approach combines a Machine Learning (ML) classifier with a crowdsourced feedback mechanism to support continuous phishing detection. The methodology is structured around three main stages: 1) feature engineering and extraction from URLs and user-reported suspicious activity; 2) classification of threats using ML models; and 3) model refinement through a crowdsourcing mechanism. This updating process aims to address the limitations of static detection systems by incorporating newly observed attack patterns. The system was evaluated on a recent phishing URL dataset, where it demonstrated strong performance in terms of precision, recall, and F1-score. The model achieved an accuracy of 99.8% under the evaluated experimental conditions. These results indicate that the proposed approach performs significantly better than conventional static detection methods tested under the same conditions. Overall, the framework provides an adaptive approach for phishing detection and highlights the potential of integrating crowdsourced feedback to support continuous model improvement.

INDEX TERMS Phishing detection, phishing fraud, URL phishing, social engineering, crowdsourcing, crowdsourced threat intelligence, dynamic detection, adaptive system.

I. INTRODUCTION

Social engineering has become a significant cybersecurity threat in virtual communities, serving as an effective method to compromise information systems. The infrastructure supporting modern work lifestyle, coupled with the pervasive use of digital communication tools, including email, instant messaging, and collaboration platforms (e.g., Skype, Dropbox, LinkedIn), has significantly amplified this risk. Notably, phishing has emerged as one of the most advanced and highly effective forms of social engineering, frequently used as the initial step to launch subsequent, more damaging attacks, such as spoofing, tampering, information disclosure, and privilege escalation [1]. The breadth and severity of this

threat are evident: phishing has impacted more than 83% of businesses, with 467,000 phishing-related emails bypassing conventional security protocols in a recent sample [2]. The financial consequences are also severe; phishing alone cost companies approximately \$17.4 billions in 2024 [2]. This widespread of phishing attack is fundamentally attributed to its ability to exploit basic human psychological principles, leveraging trust and familiarity to manipulate individuals into actions that compromise security. A very recent study showed that 30% of individuals admitted to opening suspicious emails, illustrating the enduring risk despite awareness campaigns [3].

The current landscape of phishing detection is dominated by advanced Machine Learning (ML) and Deep Learning (DL) models, with supervised approaches showing remarkable performance and high detection accuracies in

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controlled environments [4], [5], [6], [7], [8], [9], [10]. Models utilizing Logistic Regression combined with Mutual Information feature selection [4], ensemble methods like Random Forest and XGBoost [11], [12], [13], [14], and DL techniques such as LSTM [6] have demonstrated strong classification capabilities based on textual, URL, and behavioral features. However, a critical review of this state-of-the-art reveals three overarching systemic limitations that undermine their utility against the rapidly evolving threat landscape. **First**, the vast majority of high-accuracy models rely on static, small, or historically collected datasets [7], [10], [11], [14], [15]. This approach fundamentally limits their generalizability and resilience, as the models are often immediately obsolete upon deployment, struggling to detect zero-day or newly emerging phishing tactics not present in the training corpus. **Second**, existing solutions, even when implemented as browser extensions [8], lack a structured, continuous real-time retraining mechanism. Although models are accurate with known data, they cannot autonomously and efficiently incorporate new attack intelligence to update their knowledge base, rendering them passive against adversarial shifts. **Finally**, while the concept of leveraging human input has been explored to support AI in many contexts [16], current phishing detection systems fail to operationalize crowdsourcing as a primary, real-time input channel for collecting malicious URLs and immediately feeding that threat intelligence back into the ML model.

To address these aforementioned challenges, particularly the challenge of how to facilitate continuous identification of evolving phishing attacks, this research introduces a novel framework that integrates the accuracy of high-performance ML models with a dynamic, crowdsourced threat intelligence feedback loop. This system is founded on the principle that the collective, real-time experiences of users can be transformed into actionable threat intelligence, directly answering how the developed ML model can adapt to new attack tactics as they emerge. The proposed approach supports phishing detection through advanced machine learning techniques and operates across three dynamic phases: (1) Users submit suspicious URLs, from which the system automatically performs Feature Extraction. (2) Extracted data are then fed into a high-performance ML model to analyze patterns and classify the URL, effectively demonstrating how to utilize ML capabilities to detect and classify attacks. (3) Crucially, new user-reported phishing URLs are incorporated into the dataset, allowing the model to be regularly retrained without user intervention. This continuous feedback loop ensures that the system incorporates real-time user experiences and adapts to the latest attack vectors.

The proposed system integrates crowdsourced feedback with a machine learning model to enhance adaptability in phishing detection. The approach is designed to support the identification of emerging phishing patterns, including previously unseen attacks, within the evaluated dataset. Experimental results demonstrate improved detection performance compared to selected baseline methods.

The principal contributions of this research to the field of cybersecurity and machine learning-based detection are summarized as follows:

- 1) Introducing the *first* comprehensive, end-to-end system that integrates crowdsourced feedback loop directly with an adaptive Machine Learning model for real-time URL phishing detection.
- 2) Introduce a *human-in-the-loop* mechanism for identifying and incorporating newly observed phishing attacks into the training process.
- 3) Developing a sophisticated ML model capable of *adapting* to zero-day and emerging phishing tactics by seamlessly incorporating and learning from user-reported attacks.
- 4) Implementing a fully automated backend that handles the *feature extraction*, *detection*, and *retraining* of the ML model based on new user input, eliminating manual intervention.
- 5) Achieving and validating *superior detection accuracy* against recent URL datasets, demonstrably outperforming existing static and conventional benchmark systems.

The rest of this paper is organized as follows: Section II offers a concise overview of key concepts central to this research and presents the necessary background information to understand the project. Section III comprises a literature review, focusing on recent research within this field and comparing between them and highlighting distinctions and research gaps. In Section IV, we provide a detailed exposition of the system's structure, constituent elements, functionalities, and interrelationships, supported with illustrations as well as a detailed description of the experimental setup and evaluation methodology. Section V presents the results and discussions, highlighting the model's performance and a comparison benchmark analysis against existing studies. Lastly, the paper is concluded in Section VI.

II. BACKGROUND

This section establishes the technical and conceptual core foundations central to this research. Starting with phishing as the primary threat vector, the role of machine learning (ML) in its detection, and the necessity of integrating crowdsourcing for dynamic threat intelligence.

A. PHISHING ATTACKS

Phishing is the most pervasive and economically damaging form of social engineering, characterized by an attacker masquerading as a credible, trusted entity (e.g., a financial institution, a service provider, or an internal IT department) to disseminate deceptive electronic communications. These communications seek to coerce the recipient into revealing sensitive data (such as login credentials or financial information) or executing actions that compromise security [17]. Phishing attacks are fundamentally effective because they exploit inherent human psychological vulnerabilities like trust, urgency, and perceived authority, often bypassing robust

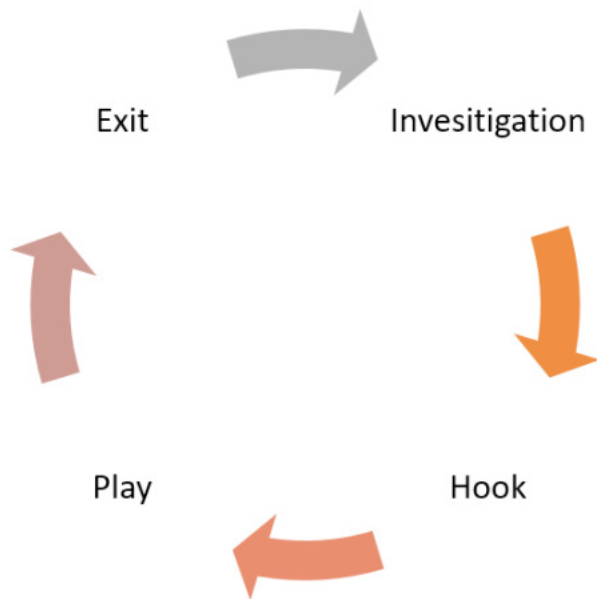


FIGURE 1. Phishing attacks life-cycle.

technological defenses. This reliance on the human element, rather than complex code exploits, establishes the user as the weakest link in the security chain [1], [18]. A generalized phishing attack follows a destructive life cycle, regardless of the channel used [18]. These steps are illustrated in Figure 1 and can be described as follows:

- 1) **Investigation/Reconnaissance:** The attacker selects a target and gathers preliminary data (e.g., contact information, organizational context).
- 2) **Lure (The Hook):** A deceptive communication is crafted to establish a false pre-text (e.g., a “security alert” or “invoice overdue”) to gain immediate trust and engagement.
- 3) **Exploitation (The Play):** The victim is manipulated into interacting with a malicious element, typically a fraudulent website link (URL), where data is harvested or malware is initiated.
- 4) **Exit:** The attacker obtains the information, terminates the communication, and evades detection.

Phishing is not limited to traditional email but transpires across numerous digital channels, diversifying into specialized attacks based on the communication medium and targeting strategy. This diversification includes Spear Phishing, which consists of highly personalized attacks aimed at specific individuals using gathered context (often through reconnaissance, similar to Pretexting); Whaling, which exclusively targets high-value executives; Smishing, which utilizes SMS text messages; and Vishing, which is conducted via Voice over IP (VoIP) or phone calls. Furthermore, attackers utilize methods like Malicious Baiting, offering false incentives (e.g., “free downloads”) to trick users into clicking malicious URLs, demonstrating that the attack vector constantly adapts across media to exploit the victim’s trust and curiosity [19].

B. MACHINE LEARNING (ML)

Machine learning (ML), a core subset of Artificial Intelligence, is utilized to overcome the limitations of static, rule-based phishing filters by dynamically learning patterns indicative of malicious activity. Its role is pivotal in processing vast amounts of data, such as URL components, textual content, and web features, to make rapid classification decisions. The success of ML-based detection systems hinges critically on the chosen algorithm, the quality and diversity of the training data, and effective feature selection [20]. Machine learning models typically work in three key phases: preprocessing, training, and testing, with feature selection often considered a crucial sub-phase. Preprocessing involves cleaning and transforming raw data handling missing values, encoding categorical variables, and scaling features to ensure it is suitable for modeling. Feature selection, which may occur during or after preprocessing, identifies the most relevant attributes from the dataset, helping to reduce noise, prevent overfitting, and improve efficiency and accuracy. After preprocessing the data, the model is trained. During the training phase, the model learns patterns from vast quantities of labeled URL and content data (supervised learning) or detects anomalies (unsupervised learning) by adjusting parameters to minimize classification error. Finally, in the testing phase, the model is evaluated on unseen data to measure its ability to generalize, using performance metrics like accuracy, precision, or recall on the task [20].

C. CROWDSOURCING FOR DYNAMIC THREAT INTELLIGENCE

Crowdsourcing is an innovative approach that leverages the potentials of artificial intelligence by collecting voluntary contributions from a large, diverse, and geographically dispersed group of individuals. Crowdsourcing is strategically applied across the ML lifecycle, particularly in data collection, model training, and validation processes. One of the most common applications of crowdsourcing in ML is data collection, where data in the datasets can be collected from a wide range of sources, making the data more representative of real-world scenarios. This method provides a mechanism to rapidly gather real-world data that surpasses the scope and latency of centralized blacklists or historically acquired security datasets [21]. By incorporating crowdsourcing in data collection, we help in providing more training data for machine learning models, as well as improving models through iterative feedback and input from a large group of users [22]. The ability of ML systems to continuously learn and generalize from new contexts makes them indispensable for developing adaptive defenses against evolving phishing tactics.

III. LITERATURE REVIEW

Phishing attacks are the most effortless approach for gathering sensitive data, such as passwords, log-in details,

and bank accounts, from unsuspecting individuals. Consequently, cybersecurity researchers are actively investigating reliable and effective procedures to detect such attacks. Researchers have explored a variety of techniques to detect phishing attacks and malicious behaviors, and ML has emerged as the dominant paradigm in these techniques. This review is structured around the primary detection methodologies: Supervised ML Approaches, Unsupervised ML Approaches, and the novel integration of Crowdsourcing with ML. In the following paragraphs, we review and explore the most up-to-date existing literature across these three categories to highlight their methodologies and limitations. This review helps to contextualize this research.

A. DETECTION USING SUPERVISED ML APPROACHES

Supervised learning techniques, which are based on labeled datasets of malicious and benign URLs or emails, currently yield the highest reported accuracy rates, where highly accurate detection is often attributed to intelligent feature selection. The work by Vajrobol et al. [4] introduced a highly efficient model for phishing URL detection that combined Mutual Information (MI) feature ranking with Logistic Regression. Achieving an exceptional accuracy of 99.97% using a highly compact subset of only five features, this study demonstrated that simple, computationally lightweight models can achieve state-of-the-art results when coupled with robust feature engineering. Despite this performance, the approach relies on manual feature engineering and lacks evaluation against real-time or evolving threats, raising concerns about overfitting to the static PhiUSIIL dataset [23]. Similarly, Etem and Teke [5] employed t-Distributed Stochastic Neighbor Embedding (t-SNE) for dimensionality reduction, reducing the feature count from 51 to 2. While the Decision Tree model achieved 99.7% accuracy on the original dataset, the notable performance degradation observed in some models after feature extraction, coupled with the reliance on a single dataset, underscores the challenge of balancing computational efficiency with information retention and generalization capacity.

Furthermore, a variety of classifiers have been employed across different data modalities. Alnemari and Alshammari [11] compared traditional classifiers for phishing domain detection, finding Random Forest (RF) to be the most effective, achieving 97.3% accuracy on the UCI dataset [24]. Taofeek [14] introduced a distinctive approach to identifying fraudulent websites by using a novel feature extraction technique: utilizing the BinVis tool to convert website source code into 2D RGB images for classification by a Convolutional Neural Network (CNN). Evaluated using features extracted from a dataset of 5,000 URLs (PhishTank and banking sources), the XGBoost model demonstrated the best performance, achieving 94.16% accuracy. While innovative, this approach shares the common limitations emphasized in other studies, including dataset constraints, which challenge its generalization capacity against the

constant evolution of phishing tactics. Fan [9] utilized ensemble learning (combining SVM and Naïve Bayes) with textual, visual, and URL features, reporting 99.16% accuracy, but acknowledged limitations regarding scalability and dependence on predefined features that evolving tactics could circumvent. In specialized applications, Butt et al. [6] focused on cloud-based email phishing, where SVM and LSTM models demonstrated strong performance (up to 99.62% accuracy), and Rathee and Mann [7] reviewed the literature showing RF implementations reaching 99.87% accuracy.

A critical challenge remains in real-world deployment. Several studies have attempted to translate high accuracy into usable tools. Priya et al. [10] developed an RF/LR-based browser extension (97% accuracy) on a modest dataset of 11,055 URLs, illustrating the feasibility of client-side implementation, but neglecting the discussion of real-time performance, latency, and computational overhead essential for practical deployment. Sawant et al. [8] integrated ML/DL and NLP techniques, including a custom CNN, to create a browser extension; however, the lack of a thorough evaluation in high-traffic, dynamic environments and concerns about overfitting and usability were noted as critical shortcomings. Ultimately, a common limitation that spans nearly all supervised work is the reliance on static, historically collected datasets [7], [11]. This constraint fundamentally limits their ability to identify zero-hour phishing attempts or adapt to unexpected attack traits [7], [9]. Furthermore, performance in non-phishing specific domains, such as the general social engineering detection by Wang et al. [15] (DT F1-score: 87.6% on only 225 samples) or the NLP-based social engineering detection by Lopez and Camargo [12] (SVM accuracy: 84.1%), highlights that generalized models struggle with the high accuracy demanded for phishing defence. This challenge extends to related textual security domains, as demonstrated by the fake news detection work of Nirmala and Navya [13], where the model reliance on training data quality and potential dataset bias constrained generalization against emerging disinformation tactics.

B. DETECTION USING UNSUPERVISED ML APPROACHES

Unsupervised learning offers a mechanism to counter the static nature of supervised models by focusing on anomaly detection rather than classification based on known labels. Assefa and Katarya [25] proposed an Autoencoder deep learning model for URL phishing detection. By employing anomaly analysis on 16 distinct features, the Autoencoder achieved an accuracy of 91.24%, outperforming comparative SVM and DT models. This approach is valuable because it is inherently suited to identifying zero-hour or novel phishing websites that deviate significantly from the baseline of legitimate sites. However, the reported accuracy, while promising for an unsupervised method, remains lower than peak supervised models, and the model's effectiveness is constrained by its relatively limited dataset size.

C. MERGING CROWDSOURCING WITH ML APPROACH

The concept of integrating large-scale human input to improve automated systems has been explored in computational fields, providing the conceptual foundation for this research. Deng et al. [16] discussed the integration of crowdsourcing into the auditing of Generative AI (GenAI) systems, emphasizing the potential of non-expert crowds to detect biases and vulnerabilities overlooked by developers. This study established a critical research agenda centered on building responsible crowdsourcing systems. Although relevant, this conceptual work lacks empirical validation and practical application of its ideas in a real-time defense context. Crucially, existing security literature largely fails to operationalize crowdsourcing as a continuous, live threat intelligence input channel for ML detection models. The principal challenges identified in this domain—including maintaining input integrity, incorporating feedback into development workflows, and establishing psychological safety for contributors—must be overcome to effectively leverage crowdsourcing for dynamic phishing detection.

In conclusion, this section provided an in-depth review of current solutions in phishing detection, dominated by high-accuracy but static supervised ML and DL techniques. The limitations and achievements of the reviewed literature are systematically documented in Table 1, which summarizes existing approaches and highlights the limitations addressed by the proposed method. These limitations converge into three critical research gaps. First, the dependence on static datasets is observed in most literature, characterized by an over-reliance on historically collected data and benchmark repositories (e.g., PhiUSIIL [23], UCI [24]), fundamentally limits the generalizability and practical effectiveness of existing models against the continuous, zero-day evolution of phishing tactics. Second, there is a clear absence of dynamic adaptation, as current solutions fundamentally lack a proven, automated mechanism for continuous retraining and refinement using live threat intelligence. Finally, a significant failure to operationalize crowdsourcing exists, representing a gap in translating the recognized potential of collective human input into a structured, production-grade feedback loop capable of providing real-time, validated malicious URLs for immediate model updates. To address these limitations, our research proposes a dynamic framework that combines the classification of advanced ML with automated crowdsourced threat intelligence feedback loop, delivering adaptive solution to evolving phishing threats.

IV. SYSTEM DESIGN AND METHODOLOGY

This section presents the architecture and methodology of the proposed phishing detection system. The system integrates machine learning with a dynamic crowdsourcing mechanism to enable adaptive detection of evolving phishing threats.

A. SYSTEM DESIGN

The system is structured into a sequential, three-phase framework as illustrated in Figure 2. The figure shows these three main phases: Extractor, Detection, and Updating. Each phase is responsible for a specific task in the processing pipeline. This modular design ensures clear separation of responsibilities between the feature extractor, the detection model, and the crowdsourced update mechanism.

1) EXTRACTOR PHASE

In the *Extractor Phase*, the input URL is processed to extract key features such as domain, TLD, and structural attributes. If the URL matches a known attack pattern, the corresponding result is returned directly. Otherwise, the extracted features are forwarded to the Detection Phase.

2) DETECTION PHASE

In the *Detection Phase*, the trained ML model is used to analyze the extracted features in order to detect if this new record is phishing or not. If the input does not match a known pattern, it is flagged as a potential novel attack and passed to the updating phase.

3) UPDATING DETECTION BY CROWDSOURCING PHASE

The *Updating Detection by Crowdsourcing Phase* provides the mechanism for adaptive defense. When the system flags an input as a novel (unknown) attack, the URL and its features are queued for human verification (typically by an administrator or security expert), which is essential for filtering out noisy or low-quality user-submitted reports. Once verified and labeled, this new record is added into the training dataset. As multiple users report similar emergent threats, a powerful crowdsourced collection of new threats is rapidly amassed. The *ML Detection Model* is periodically retrained with this expanded dataset, allowing it to remain highly adaptive and current. This continuous learning cycle directly addresses the pervasive limitation of static dataset dependence identified in the literature, significantly enhancing the model's generalization capacity against new phishing tactics.

B. METHODOLOGY AND EXPERIMENTAL SETUP

The experimental methodology is rigorously defined in this section, encompassing the dataset selection, the necessary preprocessing steps to mitigate bias and optimize data quality, and the architecture of the proposed ML model used for training and detection.

1) DATASET

Our empirical evaluation utilizes the PhiUSIIL Phishing URL Dataset [23], a comprehensive and highly relevant public resource. This dataset comprises 235,804 URLs (134,850 legitimate and 100,945 phishing URLs) and is defined by 54 structural, lexical, and content-based features extracted from both the URL and webpage source code. Given its

TABLE 1. Comparison between reviewed work in social engineering detection.

Paper / Authors	Focus Area	Key Techniques	Accuracy	Dataset	Limitations
Vajrobol et al. [4]	Phishing URL detection using feature selection and logistic regression	MI, Logistic Regression	99.97%	PhiUSIIL	Heavy feature engineering, risk of overfitting, static data dependence
Etem & Teke [5]	Phishing URL detection with dimensionality reduction	t-SNE, DT, kNN, LR, NB, RF	DT: 99.7%, kNN: 99.2%	PhiUSIIL	Performance degradation with feature reduction, static data, limited DL exploration
Alnemari & Alshammari [11]	Phishing domain detection	ANN, SVM, DT, RF	RF: 97.3%	UCI phishing dataset	Static dataset, scalability concerns, limited real-world generalization
Agboola [14]	Phishing detection via feature extraction/visualization	XGBoost, RF, CNN-based binary visualization	XGBoost: 94.16%	5,000 phishing URLs (PhishTank & banks)	Generalizability issues, controlled environment, static data
Fan Zichen [9]	Detection/classification of phishing websites	SVM, Naive Bayes, Dynamic Markov Chain	99.16%,	PhishTank (21,615 URLs)	Reliance on predefined features, limited generalization, lacks practical deployment discussion
Butt et al. [6]	Phishing email detection (cloud)	SVM, LSTM, Naive Bayes	SVM: 99.62%, LSTM: 98%, NB: 97%	Public email datasets	Limited dataset, lack of real-world testing, limited model explainability
Rathee & Mann [7]	Review of ML/DL for phishing email detection	SVM, RF, LR, NB, CNN, RNN	RF: Up to 99.87%	Various static datasets	Reliance on static datasets, proneness to false positives, manual feature engineering focus
Priya et al. [10]	Phishing detection browser extension	RF, Logistic Regression	RF: 97%	11,055 URLs	Limited dataset diversity, lack of real-time performance evaluation, limited scalability
Sawant et al. [8]	Integrated ML and DL for phishing detection	RF, XGBoost, LightGBM, Custom CNN	CNN: 95%; RF: 96.6%	Kaggle datasets (500,000 URLs)	Dataset diversity issues, no usability testing, overfitting, no dynamic evaluation
Wang et al. [15]	General social engineering detection	DT, RF, SVM, Knowledge graph approach	DT: 87.6% (F1-score)	225 samples	Very small dataset, poor generalization, complexity/interpretability
Lopez & Camargo [12]	Social engineering detection using NLP	NLP, SVM, NN, RF	SVM: 84.1%	Synthetic and real text dialogues	Small dataset, language restriction (Spanish), low accuracy for critical use
Nirmala & Navya [13]	Fake news detection	NLP, NB, Logistic Regression, DT, RF, SVM	LR: 98.84%, SVM: 99.4%	News article datasets	Dataset bias, limited adaptation to new fake news, limited DL exploration
Assefa & Katarya [25]	Phishing detection with autoencoder	Autoencoder, DT, SVM	Autoencoder: 91.24%, DT: 86.1%, SVM: 88.4%	10,000 URLs	Limited dataset size, generalization concerns

size and recent publication date (2024), the PhiUSIIL dataset is utilized to demonstrate our model's capacity to handle extensive data and cover the latest URL attack variations, addressing the problem of limited dataset size common in related literature.

2) PREPROCESSING AND MODEL TRAINING DATA PREPARATION

Cleaning and preparing the dataset are fundamental steps to ensure optimal model performance and mitigate bias.

Our preprocessing pipeline rigorously handles categorical attributes, noisy features, and class imbalance. Specifically, the methodology employed encompasses the following critical steps:

- 1) **Domain and Text Cleaning:** The “www.” prefix is removed for uniformity. Custom functions clean textual attributes (URL and Title) by removing special and non-ASCII characters. Only the main domain name (excluding TLD and subdomains) is retained for focused analysis.

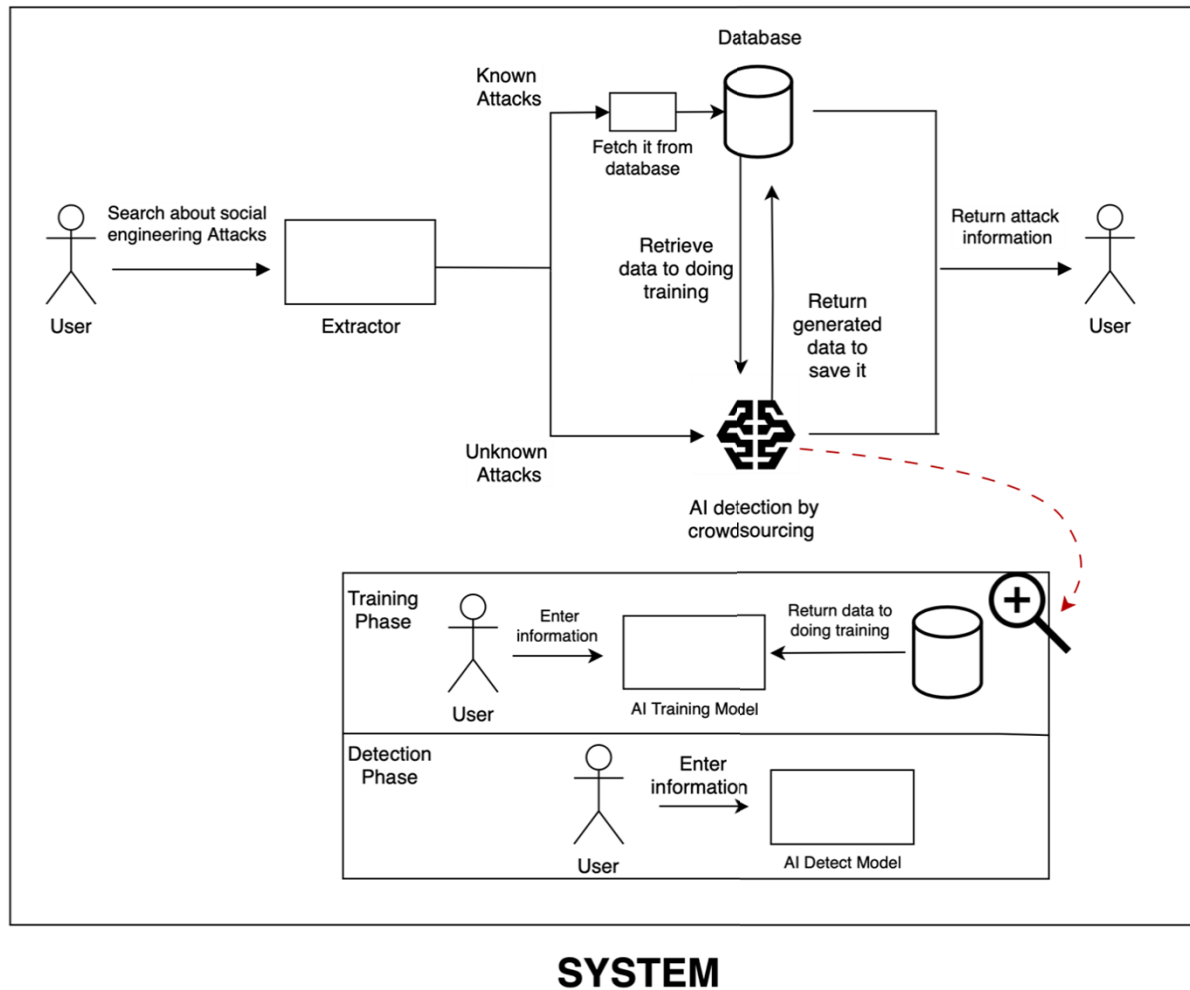


FIGURE 2. The proposed phishing detection system design.

- 2) **Data Integrity:** Duplicate rows and instances with missing values in the critical Label column are systematically dropped to maintain training integrity and data quality.
- 3) **Feature Transformation:** Categorical features are converted into numerical representations using **One-Hot Encoding** to facilitate effective learning by the ML architecture.
- 4) **Feature Selection and Engineering:** A two-stage statistical approach is employed to retain only the most discriminative features:
 - **Analysis of Variance (ANOVA) Test:** Used to evaluate the statistical significance of continuous features' impact on the Label. Features with a low F -statistic or a high P -value are excluded.
 - **Chi-Square Test:** Used to measure the correlation between categorical features and the Label.
 Features determined to be ineffective by these tests, along with manually identified noisy columns (e.g., URLLength, DomainLength), are removed.

- 5) **Handling Class Imbalance via Undersampling:** The initial class distribution reveals an imbalance in the counts of the two classes ($Counter\{1 : 134, 850, 0 : 100, 854\}$). In sensitive domains like fraud detection, this bias towards the majority class (Class 1) can lead to high apparent accuracy but poor detection performance for the minority class (Class 0). To ensure fair learning, the `RandomUnderSampler` from the `imblearn` library is applied to reduce the majority class samples, achieving a perfect balance ($Counter\{0 : 100, 854, 1 : 100, 854\}$). This process mitigates model bias, resulting in a more realistic and reflective measure of true performance across both classes.

The rigorous preprocessing and feature selection process ensures that only the most relevant, non-redundant features are retained for model training. The final selected feature set contains 34 features.

3) ML MODEL ARCHITECTURE AND TRAINING

Our model is a hybrid Deep Learning architecture designed to process both sequential text data (URL strings) and

structured numerical features concurrently. Sequential text data refers to features like the raw URL string and its Title—information where the order and context of characters and words are critical for classification. Structured numerical features refer to the computed, discrete attributes derived during the extraction phase (e.g., IsHTTPS, NoOfSubDomain, TLDLegitimateProb), where positional order is not inherently meaningful. The design, which leverages the strengths of recurrent layers for text analysis and dense layers, highlights the parallel processing of textual and numerical features for classification and is visually represented in Figure 3. This architecture integrates Long Short-Term Memory (LSTM) layers with an Attention mechanism to capture linguistic context, while processing the supplementary numerical features in parallel to achieve a comprehensive classification model.

Text Data Processing Path (Sequential Features): This path starts with an Embedding layer (50 dimensions) that converts the input URL sequence into dense numerical vectors. These vectors are then processed by a Long Short-Term Memory (LSTM) layer designed to capture temporal and linguistic dependencies within the sequence. Following this, an Attention layer is dynamically applied, enabling the model to assign higher weights to the most critical portions of the URL (such as suspicious keywords or long paths) for enhanced text representation. Finally, a second LSTM layer compresses this rich, context-aware representation into a 64-dimensional vector.

Numerical Feature Processing Path (Structured Features): The 34 preprocessed numerical features are fed through a separate Dense layer that transforms the input into a 64-dimensional vector, matching the output dimension of the text path.

Concatenation and Classification: This final phase of the architecture begins when the outputs of the two parallel branches (text and numerical processing) are merged via a Concatenate layer, creating a unified 128-dimensional feature vector. This comprehensive representation is then passed through a final Dense layer that produces the single-dimensional output, representing the calculated probability of the URL belonging to the phishing class (binary classification).

Training Strategy: The balanced dataset is split into training and testing sets. Crucially, we employ the `stratify=y_resampled` parameter during the split. This ensures that the equal distribution of classes achieved during undersampling is precisely maintained across both the training and test sets. This stratified approach is essential as it forces the model to learn true class differentiation, ensuring the final accuracy metrics are realistic and reflect the model's true generalized performance. The training process, particularly the initial feature extraction and the iterative deep learning experimentation, required significant computational time due to the large dataset size.

4) EVALUATION CRITERIA AND CROSS-VALIDATION

Evaluation metrics are indispensable tools for assessing the performance and reliability of Machine Learning models in critical security applications. To provide a comprehensive and unbiased assessment of our model's effectiveness, we utilized the following statistical metrics, defined by the fundamental concepts of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN):

- 1) Accuracy: Measures the overall correctness of the model by calculating the ratio of correctly predicted instances (TP+TN) to the total number of predictions (TP+TN+FP+FN).
- 2) Precision: Measures the quality of the positive predictions made by the model. It is the ratio of correctly predicted positive observations (TP) to the total predicted positive observations (TP+FP).
- 3) Recall (Sensitivity): Measures the model's ability to correctly identify all actual positive instances. It is the ratio of correctly predicted positive observations (TP) to all actual positive observations (TP+FN).
- 4) F1-Score: The F1-Score is the harmonic mean of Precision and Recall. It provides a balanced measure of performance, which is particularly useful in binary classification problems like ours, where maintaining high detection ability (Recall) while minimizing false alarms (Precision) is critical.

To ensure the reported performance is robust and not influenced by a single data partition, we employed 5-fold cross-validation during the model validation phase. The dataset is systematically divided into five subsets; the model is trained on four subsets and validated on the fifth, repeating this process five times until every subset has served as the validation set. The average performance across all five iterations is reported to ensure an unbiased measure of generalization capability.

Finally and in summary, the system operates through a three-phase workflow: extraction, detection, and updating. The process begins with extracting key features from the input URL, followed by classification using the trained ML model. Uncertain or novel instances are flagged and processed through a crowdsourced verification mechanism. The validated data is then integrated into the training dataset, allowing periodic model retraining and continuous adaptation to evolving phishing patterns.

V. RESULTS AND DISCUSSION

The model underwent rigorous training and evaluation on the balanced, stratified test set derived from the PhiUSIL dataset. The training process optimization and the resultant classification performance are summarized visually in Figure 4. The figure shows the training and validation performance, indicating stable learning behavior and minimal overfitting.

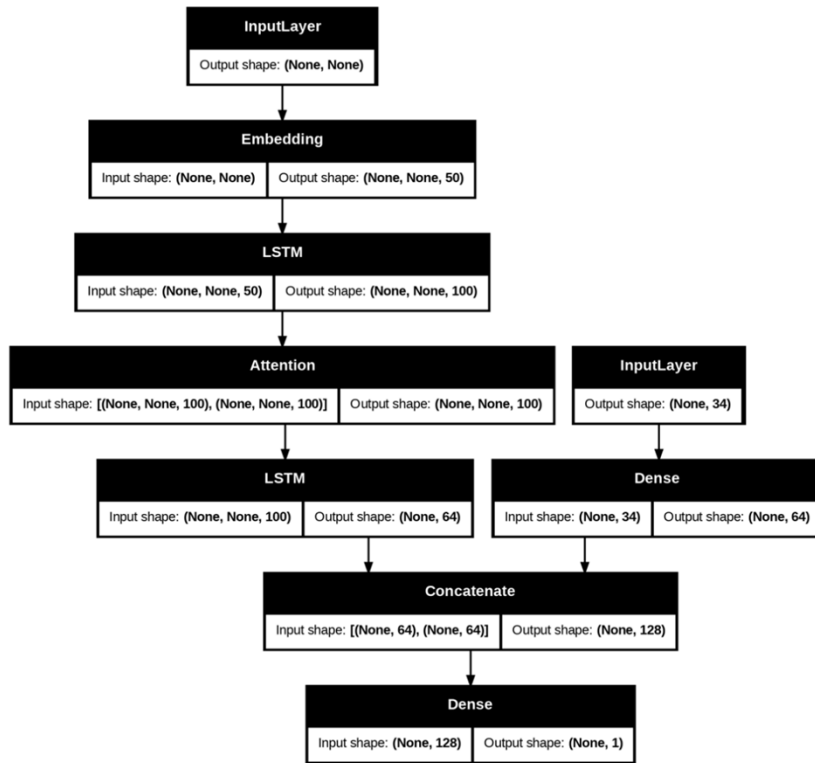


FIGURE 3. ML detection model architecture.

The final 0.9980 test accuracy confirms the model generalized effectively to unseen data, as this value is higher than the average validation accuracy during training (≈ 0.996). Moreover, the Validation Loss remained low throughout the training (Epochs 2 – 8) and the final test loss (0.0262) is comparable to the best validation losses, suggesting the model did not suffer from significant overfitting.

These results obtained confirm the effectiveness of our methodology. This high performance is attributed not only to the power of the hybrid DL architecture (integrating LSTM and Attention for sequential data) but, more importantly, to the rigorous preprocessing and data engineering applied:

- **Feature Selection:** The two-stage statistical filtering (ANOVA and Chi-Square) ensured that the model trained only on the most discriminative features, maximizing the separability of the two classes.
- **Class Balance:** The application of RandomUnderSampler eliminated class bias, ensuring high recall and precision figures and reflect the model's true differentiation ability across both classes. The stratification of the test set further validated this generalized performance.
- **Zero-Day Adaptability:** The hybrid model's ability to process complex features like URL text and domain structure demonstrated high effectiveness, highlighting that reliance on single, basic features (e.g., IsHTTPS) is insufficient. Our observation confirmed that even URLs beginning with "https" were correctly classified

TABLE 2. Detailed results of the proposed system.

Accuracy	Precision	Recall	F1-score
0.9988	0.9993	0.9983	0.9988

as phishing based on deeper structural and lexical indicators.

A detailed breakdown of the model's performance on the test set, reflecting the 0.9988 F1-score and overall accuracy, is presented in Table 2. This table provides the detailed performance metrics, confirming the model's effectiveness across all evaluation measures.

The metrics demonstrate exceptional model performance across both classes. Specifically, the near-identical Precision and Recall values for both Class 0 and Class 1, resulting in a stable 0.9988F1-score, underscore the successful mitigation of class imbalance bias through the RandomUnderSampler technique.

A. CROWDSOURCING PERFORMANCE ANALYSIS

The core novelty of this research is the integration of a dynamic, crowdsourced intelligence loop designed to continuously add the training dataset with new threat intelligence. As true, long-term operational validation of crowdsourcing involves ethical and input-integrity challenges, we conducted a learning curve analysis to simulate the effect of continuous data acquisition on model performance.

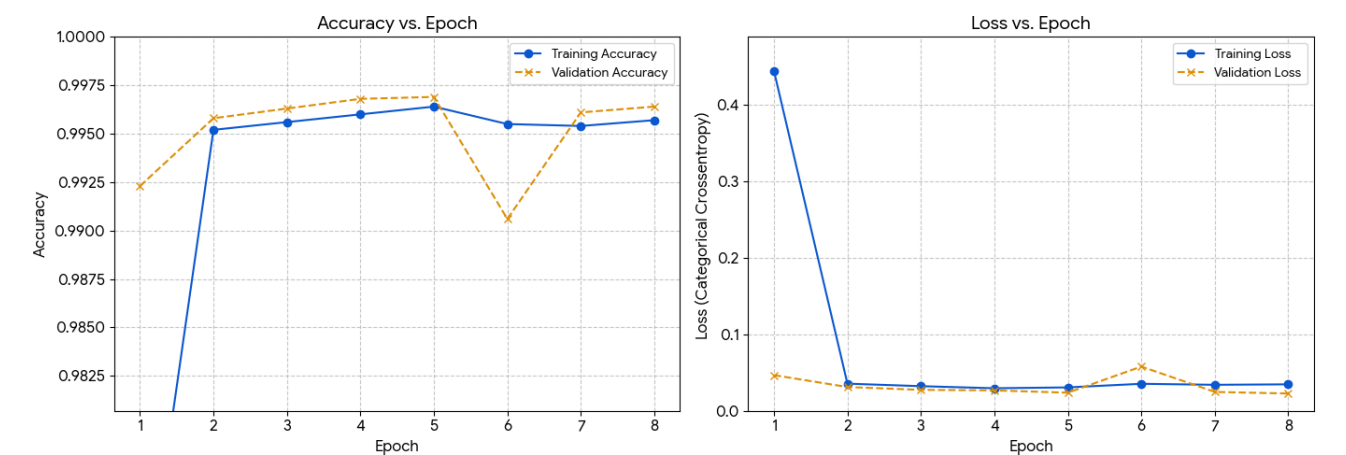


FIGURE 4. Detection model performance: Accuracy and loss curves.

This experiment involved a controlled loop where the model was trained iteratively, starting with 10% of the available training data and incrementally increasing the training set size by 10% in each step, up to the full 100%. The model was evaluated against the same, fixed independent test set at each step to observe how the accuracy stabilized as a function of training data quantity.

The visualization in Figure 5 clearly demonstrates the model’s performance stability as the simulated volume of crowdsourced data increases, strongly supporting the feasibility and robustness of the dynamic update mechanism. All curves achieved strong initial performance (above 99.3% for all four metrics) and plateaued near 99.7% in accuracy as the training data size surpassed 50%. Specifically, the stability of the F1-score (Figure 5.d) across all data proportions confirms that the model maintains an optimal balance between minimizing false positives (Precision, Figure 5.b) and minimizing missed threats (Recall, Figure 5.c), even with intermittent data batches. This rapid convergence and stability validates that the model’s architecture is highly effective in maximizing performance as its knowledge base grows, which is the primary value proposition of the continuous crowdsourced update mechanism. It also suggests that the model is capable of assimilating and correctly classifying recent phishing links that would otherwise bypass static blacklisting approaches.

B. BENCHMARK COMPARISON

To contextualize our results, we benchmarked our approach against existing literature that utilized the same dataset for URL-based phishing detection. Table 3 compares the proposed model with existing methods, showing improved performance under the same dataset (PhiUSIIL dataset).

The results clearly indicate that our integrated hybrid model, combined with meticulous feature and data handling, surpasses all examined studies in all reported performance metrics, establishing a new state-of-the-art accuracy

TABLE 3. Benchmark comparison.

Paper	Accuracy	Precision	Recall	F1-score
Vajrobol et al. [4]	99.4%	99.3%	99.3%	99.3
Etem & Teke [5]	99.6%	99.6%	99.6%	99.5
Our Proposed Model	99.88%	99.93%	99.83%	99.88

benchmark for detection on this dataset. This improved performance validates our core hypothesis: that rigorous data preprocessing and the architectural synthesis of sequential and structured feature analysis which mirrors the dynamic nature of phishing attacks, are key to achieving high classification efficacy in modern cybersecurity defense systems. It is important to note that the reported results are based on the evaluated dataset and experimental setup. While the model was assessed using the same experimental conditions as the benchmarked studies, further validation in real-world environments is still required.

C. COMPUTATIONAL COMPLEXITY ANALYSIS

To evaluate the efficiency of the proposed system, both theoretical and empirical analyses of time and space complexity are considered.

1) TIME COMPLEXITY

The overall time complexity of the system depends on three main stages: feature extraction, model inference, and model updating.

During the feature extraction phase, the complexity is proportional to the length of the input URL, resulting in a time complexity of $O(n)$, where n represents the number of characters in the URL. For the detection phase, the complexity is dominated by the LSTM-based model. Given a sequence length T and hidden dimension H , the time complexity is approximately $O(T \cdot H^2)$, due to recurrent computations within the LSTM layers. The crowdsourcing

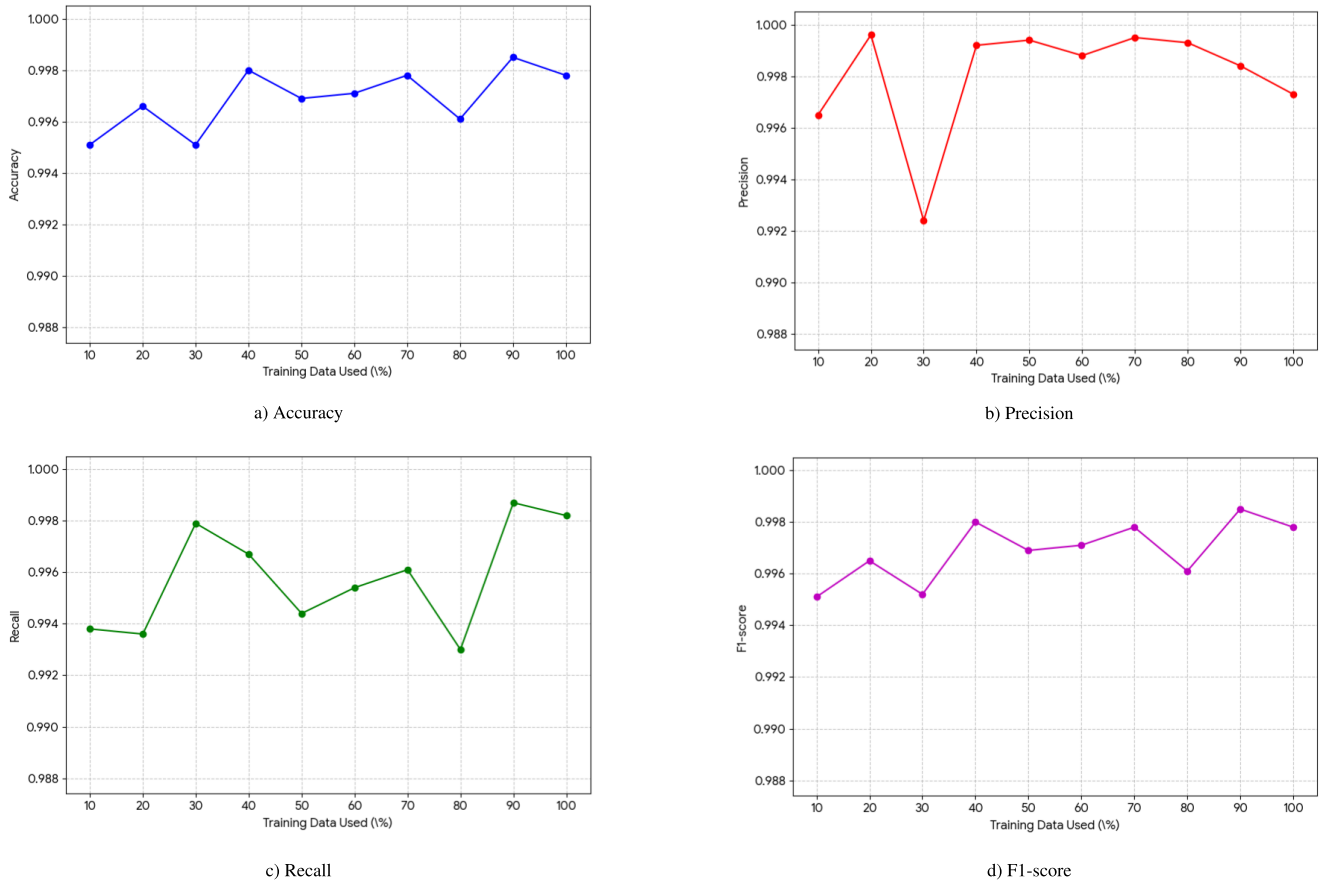


FIGURE 5. Performance stability during simulated crowdsourced data growth.

update mechanism introduces additional overhead during periodic retraining. However, this process is performed offline and does not affect real-time detection latency.

From an empirical perspective, the system demonstrates efficient real-time performance during inference, as feature extraction and prediction are performed within milliseconds per input. To quantify this, the prediction time was measured across 10 independent runs, each processing 40,342 instances. The total inference times recorded were: 8.16s, 10.31s, 7.94s, 7.96s, 8.00s, 6.65s, 7.46s, and 8.39s. These results correspond to an average inference time of approximately *0.20 milliseconds per URL*, confirming the system's suitability for real-time detection scenarios.

2) SPACE COMPLEXITY

The space complexity is primarily determined by the storage of model parameters and feature representations. The LSTM model requires memory proportional to $O(H^2)$, while the input features require $O(n)$ space. In practice, the deployed model has a size of *48 bytes*, making it highly suitable for lightweight and resource-constrained environments. This minimal model size proves that our system introduces only negligible memory overhead.

VI. CONCLUSION AND FUTURE WORK

This research aims to reduce the critical limitations of contemporary static detection methods by developing a dynamic phishing URL detection system that integrates a high-performance Deep Learning (DL) classifier with a real-time crowdsourced threat intelligence feedback loop. The proposed approach demonstrates promising results in adapting to evolving phishing patterns within the evaluated dataset. Empirical results validated the efficacy of this approach, achieving a state-of-the-art 99.8% overall accuracy on the PhiUSIIL dataset, significantly surpassing established benchmarks [4], [5]. The findings affirm the critical role of integrating collective human intelligence with automated systems to build a scalable and resilient defense against evolving fraud. Despite the advantages of crowdsourcing in improving model adaptability, it is important to consider the potential risks associated with unverified user inputs. Recent studies have shown that crowdsourced cyber threat intelligence may include misleading or false information, which can negatively impact model performance if not properly validated [26]. Furthermore, human-in-the-loop approaches have been identified as essential in high-stakes AI systems to ensure reliability and effective oversight [27]. Therefore, incorporating administrator verification mechanisms,

as outlined in the system proposed in this paper, is essential to guarantee data reliability. Looking forward, future work will focus on several critical areas, including: investigating the system scalability and computational efficiency in real-world deployment scenarios, particularly for large-scale, enterprise-level deployment. Additional directions include developing robust countermeasures against the potential adversarial misuse of the adaptive system; expanding the crowdsourcing component to include multimodal inputs and address multilingual texts; and formalizing protocols for user data anonymity and exploring XAI techniques to enhance transparency and address the ethical challenges associated with algorithmic bias and false positives.

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